

Technologies Used in SoloLLM (Roman Urdu)

Yeh document SoloLLM project mein istemal hone wali har technology, library aur framework ki tafseeli wazahat karta hai. Humne is project ko highly optimized, local AI execution aur modern web standards par banaya hai. Yeh document aapko batayega ke kaunsi cheez kyun aur kahan use ki gayi hai.

1. Backend Technologies (Python)

Backend poora Python par mabni hai kyunke AI aur machine learning ecosystem Python mein sab se zyada mazboot hai.

Core Frameworks & Servers

- **FastAPI** (`fastapi==0.115.6`): Yeh ek modern, fast (high-performance) web framework hai Python ke liye. Ise API endpoints (jaise `/chat`, `/models`, `/agents`) banane ke liye use kiya gaya hai. Yeh Pydantic ke sath mil kar data validation aur auto-documentation (Swagger UI) provide karta hai. Mukhaliq files: `backend/main.py`, `backend/api/*.py`.
- **Uvicorn** (`uvicorn[standard]==0.34.0`): Yeh FastAPI backend ko run karne ke liye as a lightning-fast ASGI server use hota hai.
- **AioSQLite** (`aiosqlite==0.20.0`): Project ka sara local data (chat history, settings, agents memories) persist karne ke liye SQLite use hota hai. AioSQLite use karne ka maqsad yeh hai ke database queries asynchronous hon aur backend block na ho. Mukhaliq file: `backend/storage/database.py`.

AI, Machine Learning & RAG Components

- **Ollama Manager**: Local LLM models (jaise Llama 3, Mistral) ko run karne ke liye project background mein Ollama ka istemal karta hai. Ollama runner khud directly unki official website (ollama.com) ya GitHub releases se download hota hai ya user ke system pe pehle se maujood hota hai. Yeh LLMs ke sath interact karne ka sab se asaan aur optimized tareeqa deta hai.
- **PEFT** (`peft>=0.13.0`) & **TRL** (`trl>=0.12.0`) & **Accelerate**: Yeh HuggingFace libraries hain jo QLoRA (Quantized Low-Rank Adaptation) fine-tuning ke liye use hoti hain. `peft` memory efficient fine-tuning (LoRA adapters) ke liye, `trl` reinforcement learning/SFT

(Supervised Fine-Tuning) ke liye, aur `accelerate` multi-GPU/hardware memory management ke liye use hota hai. Mukhaliq file: `backend/core/finetuner.py`.

- **Sentence-Transformers** (`sentence-transformers==3.4.1`): Text ko 512 ya 768-dimensional numerical vectors (embeddings) mein convert karne ke liye use hota hai. Jab bhi user koi PDF document upload karta hai, toh yeh library background mein HuggingFace se embedding model (jaise `all-MiniLM-L6-v2`) download karti hai aur text ko vector mein tabdeel karti hai.
- **FAISS** (`faiss-cpu==1.10.0`): Meta (Facebook) ki taraf se banai gayi library hai jo lakho vectors ke darmiyan similarity search instantly perform karti hai. RAG (Retrieval-Augmented Generation) pipeline mein memory search karna aur accurate documents dhoondne ke liye FAISS istemal hua hai. Mukhaliq file: `backend/rag/vectorstore.py`.
- **NetworkX** (`networkx==3.4.2`): Knowledge Graph banane ke liye use hota hai. Entities (nodes) aur unke relations (edges) ko store aur visualize karne ki mathematically routing is library ke through hoti hai. Mukhaliq file: `backend/memory/knowledge_graph.py`.

Utilities & Other Libraries

- **Pydantic** (`pydantic==2.10.4`): Data parsing aur validation ke liye. Frontend se aane wale JSON payloads ko strict Python Objects mein convert karta hai. Mukhaliq file: `backend/storage/schemas.py`.
- **BeautifulSoup4** (`beautifulsoup4==4.13.3`): Web scraping ke liye. Jab user kisi website ka URL data ingest karne ke liye deta hai, toh yeh HTML ko parse kar ke useable text banata hai. Mukhaliq file: `backend/rag/scrapper.py`.
- **NumPy** (`numpy==2.2.1`): Fast mathematical computations (vector distances, arrays) ke liye.
- **SSE-Starlette** (`sse-starlette==2.2.1`): Server-Sent Events (SSE) streaming ke liye. Jab LLM ek ek word (token) generate kar raha hota hai, toh us waqt real-time typing effect dikhane ke liye yeh library chunk-by-chunk data frontend ko bhejti hai.
- **GPUtil / psutil**: Hardware metrics (RAM, GPU usage, VRAM) check karne ke liye taake LLM crash hone se bache aur token budget manage ho. Mukhaliq file: `backend/core/hardware.py`.

2. Frontend Technologies (Next.js / React)

Frontend ko ek premium, snappy aur highly interactive web application banane ke liye in tools ka intekhab kiya gaya hai.

- **Next.js (v16.1.6)**: React ka framework jo routing (App Router), file-system based routing aur optimized build ke liye use hua hai. `frontend/src/app` mein iske sab routes define hain. Next.js SEO aur performance (server components) ke mamlay mein behtareen hai.
- **React (v19.2.3)**: UI components banane ke liye core library. State management (hooks jaise `useState`, `useEffect`) se chat interfaces aur dashboards banaye gaye hain.
- **TailwindCSS (v4)**: Modern utility-first CSS framework. Humne custom CSS files likhne ke bajaye in-line classes use kar ke UI design kiya hai, jo ke fast aur responsive (mobile-friendly) layout banata hai. Is se "glassmorphism" aur dark mode themes implement karna bohot asaan raha.
- **Lucide React (lucide-react==0.577.0)**: Minimal aur khoobsurat SVG icons ka package. User interface (AI avatars, settings gear, send buttons) mein jo bhi icons nazar aatay hain, wo yahan se render hotay hain.
- **React Markdown**: LLM models mostly Markdown text generate karte hain (bold, code blocks, tables). Is library ke zariye hum us markdown text ko properly formatted HTML components mein convert kar ke screen par dikhate hain.

3. Background Models & Downloads

Project chalne ke dauran kuch cheezein background mein download hoti hain: 1. **Ollama Binary**: First time chalane par ya system setup mein, standard Ollama client install hota hai taake Llama, Phi, aur Mistral models as local servers chal sakein. 2. **HuggingFace Embedding Models**: `sentence-transformers` automatically first run par embedding models (e.g. `openai-embed-text` ya `all-MiniLM-L6-v2`) cache mein download karta hai, taake phir hamesha offline kaam kare. Inka size taqreeban 100MB-300MB hota hai. 3. **Local LLMs (via Ollama)**: User jab UI se koi model select karta hai, toh Ollama background mein usay pull (download) karta hai jiska size 2GB se 8GB tak ho sakta hai.

Yeh sab isliye banaya gaya hai taake pura ai system offline, private aur aapke apne hardware par maximum efficiency ke sath kaam kar sake!